

Decentralized Swarm Resilience Under Stochastic Gamma Noise: A Computational Model for Pressure Vessel Inspection

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Abstract—Robotic inspection of the primary coolant loop in Pressurized Water Reactors (PWRs) is technologically underserved, owing to high-Reynolds-number turbulent flow coexisting with intense, spatially heterogeneous gamma radiation. Tethered ROVs and centralized swarms both fail systematically here: tethers suffer shear, entanglement, and radiation-induced optical attenuation, while centralized command presents a single point of failure vulnerable to dose-driven latch-up. This paper proposes a decentralized, communication-degradation-tolerant swarm framework for resilient path-planning and defect localization, instantiated as a Modified Particle Swarm Optimization (M-PSO) algorithm for micro-agent navigation under simultaneous hydrodynamic and radiological stress. A reduced-order simulator, released as supplementary material, couples a stochastic turbulent velocity field with agent-level hydrodynamic drag, a non-homogeneous radiation dose map driving dual-phenomenon degradation (Gaussian sensor corruption and dose-dependent packet loss), and a genuine cumulative-dose state governing permanent hardware failure. M-PSO uses neighborhood-restricted information propagation, gated on a per-agent binary channel-state variable, to preserve search coherence despite severe stochastic data loss. A Monte Carlo campaign (500 trials/condition), executed directly from the accompanying simulator, shows the decentralized swarm sustains 84.4% target-localization success (95% CI [81.0, 87.3]) under catastrophic gamma flux, versus 1.4% for a centralized baseline—a collapse traced to single-point failure of the leader’s communication hardware. A non-drag-aware canonical PSO baseline is also benchmarked and underperforms across all regimes. These results substantiate decentralization as a necessary architectural principle for resilient robotic inspection in harsh-radiation environments.

Index Terms—Harsh-environment robotics, resilient path planning, sensor-noise robustification, swarm robotics, particle swarm optimization, decentralized control, nuclear inspection, fault-tolerant communication.

I. INTRODUCTION

The primary coolant loop of a Pressurized Water Reactor (PWR) is a physically inaccessible, hazardous environment subject to mandatory in-service inspection, combining fully turbulent flow ($Re > 10^5$) with intense, non-homogeneous gamma radiation capable of degrading unshielded micro-electronics on contact. No mature, field-proven autonomous solution exists for this dual-hazard interior [1].

Existing technologies are ill-suited to it. Tethered ROVs, the incumbent underwater-inspection solution, suffer tether snagging and fatigue fracture under turbulent shear; fiber-optic tethers substituted to avoid EMI instead suffer radiation-induced attenuation (RIA), which progressively severs the channel [2]. Centralized swarm architectures fare no better: a designated leader agent is a single point of failure whose loss—through dose accumulation, latch-up, or single-event upset—terminates the entire mission, consistent with field-documented fragility of teleoperated robotics in high-radiation environments [3], and even recent multi-robot nuclear-decommissioning demonstrations retain a human-in-the-loop coordinator as an analogous single aggregation point [4].

This motivates decentralized, bio-inspired coordination, a direction consistent with the broader swarm-robotics literature’s continued shift toward decentralization [5]. Particle Swarm Optimization (PSO) [6] is attractive precisely because no single agent’s failure is catastrophic to the collective search. Each agent’s canonical velocity update is

$$v_i^{t+1} = \omega v_i^t + c_1 r_1 (p_i^{\text{best}} - x_i^t) + c_2 r_2 (g^{\text{best}} - x_i^t), \quad (1)$$

with $r_1, r_2 \sim U(0, 1)$ and the standard position update $x_i^{t+1} = x_i^t + v_i^{t+1}$. This presupposes flawless, zero-latency transmission of p_i^{best} and g^{best} —an assumption categorically violated in a radiologically active, turbulent PWR loop, where both physical and communicated agent state are subject to continuous stochastic corruption.

Scoping the contribution. Neighborhood-restricted social attraction is itself an established PSO variant [7], [8], not introduced here; nor is the finding that a no-single-point-of-failure topology degrades gracefully while leader-follower collapses a new theoretical result—it instances known properties from the distributed consensus literature [9]. This paper’s contribution is narrower: a *physically-grounded process for generating the neighborhood topology itself*, where graph edges are jointly gated by inter-agent range and a radiologically-driven, dual-mechanism (transient Bernoulli and permanent cumulative-dose) hardware-failure model, coupled to per-agent hydrodynamic drag specific to a turbulent PWR channel. The closest prior work, Ristic et al.’s decentralized search for a hazardous

source in turbulent flow [10], addresses estimation-theoretic localization under turbulent dispersion; this paper instead addresses swarm-optimization kinematics where the *communication medium itself*, not just the target's dispersion signature, is degraded by the hazard. A reduced-order, purpose-built simulator—rather than extending a general-purpose platform such as Wright et al.'s Gazebo radiation simulator [11]—was chosen to keep a $500 \times 3 \times 3$ Monte Carlo campaign tractable; a pragmatic trade-off, not a fidelity claim (Sec. IV).

The contributions are threefold:

- 1) **Hydrodynamic Agent Kinematics:** per-agent quadratic drag, $\vec{F}_d = -\frac{1}{2}\rho C_d A |\vec{v}_{\text{rel}}| \vec{v}_{\text{rel}}$ (where $\vec{v}_{\text{rel}}$ is the agent's velocity relative to the local fluid), coupled to PSO kinematics under $Re > 10^5$ flow, replacing idealized frictionless point-particle motion.
- 2) **Non-Homogeneous Radiation-Induced Degradation:** a spatially heterogeneous gamma dose field driving (i) dose-proportional Gaussian sensor noise and (ii) dose-dependent stochastic communication loss (Sec. II-C).
- 3) **Decentralized Adaptive Swarm Mechanisms:** neighborhood-restricted information exchange gated by a per-agent, radiologically-driven channel-state variable, preserving search efficiency under severe, spatially correlated dropout that collapses centralized coordination.

II. SYSTEM ARCHITECTURE AND MATHEMATICAL MODEL

Three subsystems, each derived from first principles, integrate into a unified per-timestep update: (A) the M-PSO kinematic update and communication graph; (B) turbulent hydrodynamic drag; and (C) the dual-phenomenon radiation degradation model. All updates advance in discrete steps of size Δt (the fixed simulation timestep, Table I); Fig. 1 summarizes how these compose into the loop executed by Algorithm 1.

A. Kinematic Swarm Update System

The motion of agent $i \in \{1, \dots, N\}$ is governed by an M-PSO update augmenting the canonical cognitive-social formulation with a physical forcing term from fluid-structure interaction:

$$\vec{v}_i(t+1) = w \vec{v}_i(t) + c_1 r_1 (\vec{p}_i(t) - \vec{x}_i(t)) + \chi_i(t) c_2 r_2 (\vec{g}_i(t) - \vec{x}_i(t)) + \frac{\vec{F}_d}{m} \Delta t, \quad (2)$$

with forward-Euler position update $\vec{x}_i(t+1) = \vec{x}_i(t) + \vec{v}_i(t+1) \Delta t$. The term $(\vec{F}_d/m)\Delta t$ is the principal departure from canonical PSO, embedding Newtonian drag directly into search kinematics. Following long-standing PSO convention [6], [12], the cognitive/social terms are an abstract velocity-space control input (not a physical force); drag, by contrast, is a genuine kinematic acceleration and is explicitly integrated over Δt so it combines correctly, in a forward-Euler sense, with the algorithmic terms rather than being added as a raw acceleration—precluding agents from executing arbitrarily precise trajectories in a medium that physically resists motion.

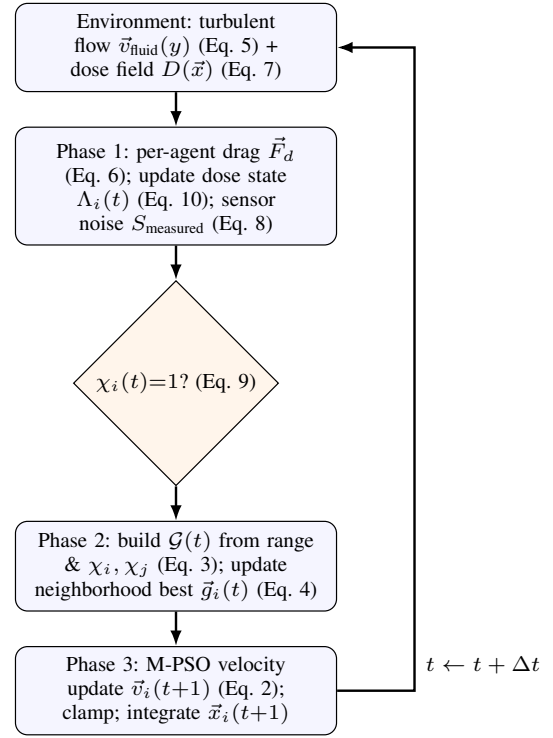


Fig. 1: Per-timestep system workflow. $\chi_i(t)=0$ (transceiver down) does not halt agent i : $\mathcal{N}_i(t)$ collapses toward $\{i\}$ and the social term of Eq. (2) is gated off, but Phase 1 and Phase 3 still execute every step for every agent.

Eq. (2) also replaces the canonical global attractor $\vec{g}^{\text{best}}$ with an agent-specific, neighborhood-local attractor $\vec{g}_i(t)$, defined next.

1) *Dynamic Communication Graph:* Let $\mathcal{G}(t) = (\mathcal{V}, \mathcal{E}(t))$ denote the swarm communication graph, $\mathcal{V} = \{1, \dots, N\}$. An edge $(j, i) \in \mathcal{E}(t)$ —a successful transmission from j to i , over a transceiver of fixed physical range r_{comm} —requires joint range and hardware operability:

$\|\vec{x}_j(t) - \vec{x}_i(t)\| \leq r_{\text{comm}}$ and $\chi_j(t) = 1$ and $\chi_i(t) = 1$, (3) where $\chi_k(t) \sim \text{Bernoulli}(1 - P_{\text{drop}}(\vec{x}_k(t)))$ is the single per-agent channel-state variable (Sec. II-C), reused for both transmit (χ_j) and receive (χ_i) roles, since both share one physical transceiver and fail together under radiation. $\mathcal{N}_i(t)$ thus already collapses to $\{i\}$ when $\chi_i(t) = 0$; Sec. II-C explains why Eq. (2) additionally gates the social term on $\chi_i(t)$ rather than relying on self-collapse alone.

2) *Neighborhood-Restricted Local Best:* The active neighborhood of agent i is $\mathcal{N}_i(t) = \{j \in \mathcal{V} \mid (j, i) \in \mathcal{E}(t)\} \cup \{i\}$, giving a decentralized, neighborhood-specific attractor

$$\vec{g}_i(t) = \arg \max_{j \in \mathcal{N}_i(t)} \{f(\vec{p}_j(t))\}, \quad (4)$$

where $f(\cdot)$ is the noise-corrupted signal fitness. Because $\mathcal{N}_i(t)$ is a stochastic function of $\mathcal{G}(t)$, $\vec{g}_i(t)$ may differ across agents and, in the limit, reduce to full isolation ($\mathcal{N}_i(t) = \{i\}$) when all inbound edges fail simultaneously.

3) *Notation:* $\vec{x}_i, \vec{v}_i \in \mathbb{R}^2$: position/velocity in the pipe cross-section. w : inertia weight. c_1, c_2 : fixed cognitive/social

coefficients, *never reassigned at runtime*—packet loss is modeled via $\chi_i(t)$, not by mutating c_2 . $r_1, r_2 \sim U(0, 1)$: exploration randomness. $\vec{p}_i(t) = \arg \max_{\tau \leq t} f(\vec{x}_i(\tau))$: personal best, under one maximization convention used consistently throughout (larger f = stronger, more target-proximal signal), including in Algorithm 1. $\vec{g}_i(t)$: neighborhood-local best (Eq. 4), evaluated over $\mathcal{N}_i(t)$, not the full swarm. $\vec{F}_d, m$: drag force and agent mass, s.t. $(\vec{F}_d/m)\Delta t$ is the drag velocity contribution per step.

B. Turbulent Hydrodynamics and Drag Modeling

Unlike the inert medium assumed by idealized PSO, the PWR coolant pipe presents a fully turbulent flow field that continuously perturbs agent motion. For fully developed flow at Reynolds number $Re > 10^5$, the cross-sectional velocity is well-approximated by the empirical one-seventh power-law profile [13] plus a stochastic fluctuation term:

$$\vec{v}_{\text{fluid}}(y) = v_{\text{max}} \left(1 - \frac{|y|}{R}\right)^{\frac{1}{7}} \hat{i} + \vec{v}_{\text{turb}}, \quad \vec{v}_{\text{turb}} \sim \mathcal{N}(0, \sigma_{\text{turb}}^2), \quad (5)$$

where $y \in [-R, R]$ is the wall-normal coordinate, R the pipe radius, v_{max} the centerline velocity, and $\hat{i}$ the unit vector along the pipe’s axial direction; the profile enforces no-slip at $y = \pm R$ and a maximum at $y = 0$. $\vec{v}_{\text{turb}}$ is a first-order stochastic representation of the Reynolds-decomposed fluctuation $u' = u - \bar{u}$, omitting full RANS closure for computational tractability while preserving the essential stochastic forcing.

This field couples to agent motion via the standard quadratic drag relation for bluff bodies:

$$\vec{F}_d = -\frac{1}{2} \rho C_d A \|\vec{v}_i(t) - \vec{v}_{\text{fluid}}(y)\|^2 \hat{u}_d, \quad (6)$$

with $\rho \approx 1000 \text{ kg/m}^3$ coolant density, C_d the drag coefficient, A the frontal area, and $\hat{u}_d$ the unit vector opposing relative velocity. This quadratic dependence is strongly non-linear: agents moving against the mean flow near the steep-gradient wall region experience disproportionate resistive forcing, biasing trajectories toward the bulk flow direction—a bias absent from idealized PSO.

C. Ionizing Radiation Degradation Fields

A spatially non-homogeneous gamma field, concentrated near the fissure (corrosion/activation sites co-locate in practice), is modeled as a background term plus inverse-square enhancement:

$$D(\vec{x}) = D_{\text{bg}} + \frac{K}{\|\vec{x} - \vec{x}_f\|^2 + 1.0}, \quad (7)$$

with D_{bg} ambient dose rate, K source intensity, and the +1.0 regularizer preventing a singularity at $\vec{x} \rightarrow \vec{x}_f$ while still permitting severe local dose—an unavoidable mission-success/survivability trade-off, since inspection requires approaching the highest-hazard region.

This drives two mechanistically distinct degradation phenomena [14]. **(a) Sensor noise:**

$$S_{\text{measured}}(\vec{x}) = S(\vec{x}) + \epsilon_\gamma, \quad \epsilon_\gamma \sim \mathcal{N}(0, \sigma_\gamma^2 D(\vec{x})), \quad (8)$$

where $S(\vec{x})$ is the true, noise-free proximity signal used as the fitness function $f(\cdot)$ throughout this paper (peaked at $\vec{x}_f$,

TABLE I: Physical and Computational Simulation Configuration Parameters

Parameter	Notation	Value
<i>Environment / Hydrodynamic Constants</i>		
Pipe Length, Radius	L, R	10.0, 1.0 m
Fissure Position	$\vec{x}_f$	(8.0, 0.9) m
Micro-agent Mass	m	0.05 kg
Coolant Density	ρ	1000 kg/m ³
Drag Coeff., Frontal Area	C_d, A	0.47, $1.26 \times 10^{-3} \text{ m}^2$
Max Fluid Vel., Turb. SD	$v_{\text{max}}, \sigma_{\text{turb}}$	1.5, 0.1 m/s
<i>Algorithmic Parameters</i>		
Swarm Size, Steps	N, t_{max}	25, 150
Timestep	Δt	0.1 s
Inertia, Cog., Social Wts.	w, c_1, c_2	0.6, 1.2, 1.2
Comm. Range, Vel. Clamp	$r_{\text{comm}}, v_{\text{clamp}}$	2.0m, 3.0m/s
<i>Radiological Hazard Constants (Gy/s)</i>		
Profile Dose Params.	D_{bg}, K	(0, 0), (.5, 10), (2, 100)
(Ctrl/Mod/Extr)		
Drop Sens., Noise Prop.	β, σ_γ	.15 s/Gy, .08
Cum. Dose Fail. Threshold	Λ_{fail}	300.0 Gy

decaying with distance), and noise variance scales linearly with local dose rate via σ_γ^2 , so fitness reliability degrades precisely as agents approach the region of greatest interest.

(b) Communication loss:

$$P_{\text{drop}}(\vec{x}) = 1 - e^{-\beta D(\vec{x})}, \quad (9)$$

with β the hardware’s dose-to-failure sensitivity ($P_{\text{drop}} \rightarrow 1$ at high dose; $P_{\text{drop}} \rightarrow \beta D(\vec{x})$ at low dose, consistent with a Poisson-thinned first-order failure process). A drop instantaneously zeroes $\chi_i(t)$ (Eq. 3), collapsing Eq. (2) to purely cognitive-inertial motion for that agent/step—informational isolation, not agent failure, since connectivity may be stochastically restored.

This transient mechanism alone cannot capture *permanent* hardware failure under sustained exposure. Each agent therefore accumulates a running dose state

$$\Lambda_i(t) = \Lambda_i(t-1) + D(\vec{x}_i(t)) \Delta t, \quad \Lambda_i(0) = 0, \quad (10)$$

and once $\Lambda_i(t) \geq \Lambda_{\text{fail}}$, the transceiver latches up permanently:

$$\Lambda_i(t) \geq \Lambda_{\text{fail}} \implies \chi_i(t') = 0 \quad \forall t' \geq t, \quad (11)$$

regardless of subsequent position, where Λ_{fail} is a fixed, hardware-specific cumulative-dose failure threshold (Table I). Only Eq. (11)’s condition is treated as permanent; ordinary Bernoulli drops via $P_{\text{drop}}(\vec{x})$ alone are always transient. It is the swarm’s aggregate resilience under this combination of recurrent transient isolation and occasional permanent hardware loss—not any individual agent’s performance—that is the central object of investigation below.

Algorithm 1 consolidates Eqs. (2)–(11) into the per-timestep loop summarized in Fig. 1. Two symbols appear only in the algorithm and Table I: $\Omega_{\text{pipe}} = [0, L] \times [-R, R]$, the pipe cross-section over which initial positions are drawn uniformly, and v_{clamp} , a fixed maximum speed applied identically to every agent and architecture—the standard PSO velocity-clamping safeguard [12], not an additional physical assumption.

III. SIMULATION CONFIGURATION AND RESULTS ANALYSIS

The model was instantiated in a discrete-time simulator (`pso_nuclear_sim.py`) executing $N = 25$ agents over $t \in [0, 150]$ steps in a 2D pipe cross-section, $R = 1.0 \text{ m}$,

Algorithm 1 Communication-Degradation-Tolerant Decentralized M-PSO

Require: N , $t_{\max}$, pipe geometry (L, R) , fissure $\vec{x}_f$; physical/flow/radiological/algorithmic constants (Table I)

Ensure: $\{\vec{g}_i(t_{\max})\}$, $\{\vec{p}_i(t_{\max})\}$

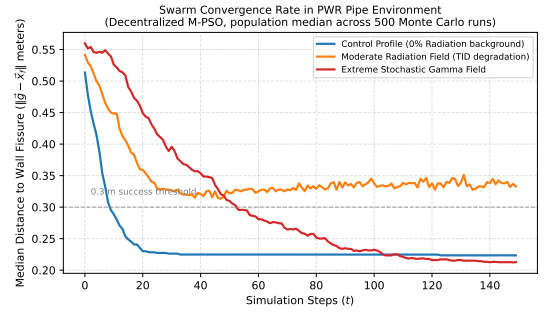
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1: for  $i = 1$  to  $N$  do
2:    $\vec{x}_i(0) \sim U(\Omega_{\text{pipe}})$ ;  $\vec{v}_i(0) \leftarrow \vec{0}$ ;  $\vec{p}_i(0) \leftarrow \vec{x}_i(0)$ ;  $\Lambda_i(0) \leftarrow 0$ ;
   evaluate  $f(\vec{p}_i(0))$ 
3: end for
4: for  $t = 0$  to  $t_{\max} - 1$  do
5:   for  $i = 1$  to  $N$  do  $\triangleright$  Phase 1: physical + radiological update
6:     Compute  $\vec{v}_{\text{fluid}}$ ,  $\vec{F}_d$  (Eqs. 5–6); update  $\Lambda_i(t)$  (Eq. 10); latch
     if  $\Lambda_i(t) \geq \Lambda_{\text{fail}}$ 
7:       Draw  $\chi_i(t)$  per Eq. 9 (force 0 if latched)
8:       Evaluate  $S_{\text{measured}}$  (Eq. 8); update  $\vec{p}_i(t)$  if improved (max-
       imization, Sec. II-A3)
9:   end for
10:  Construct  $\mathcal{G}(t)$  (Eq. 3)  $\triangleright$  Phase 2: communication
11:  for  $i = 1$  to  $N$  do
12:    Build  $\mathcal{N}_i(t)$ ; update  $\vec{g}_i(t)$  (Eq. 4)
13:  end for
14:  for  $i = 1$  to  $N$  do  $\triangleright$  Phase 3: kinematic integration
15:    Draw  $r_1, r_2$ ; update  $\vec{v}_i(t+1)$  (Eq. 2); clamp speed to
     $v_{\text{clamp}}$ 
16:     $\vec{x}_i(t+1) \leftarrow \vec{x}_i(t) + \vec{v}_i(t+1)\Delta t$ ; clip to pipe boundary
17:  end for
18: end for
19: return  $\{\vec{g}_i(t_{\max})\}$ ,  $\{\vec{p}_i(t_{\max})\}$ 

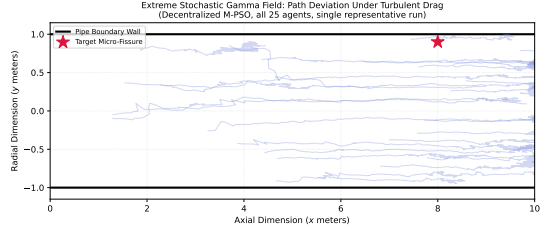
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with the target fissure near the outer wall at $(x_f, y_f) \approx (8.0, 0.9)$ m—a deliberately near-boundary placement within the steepest turbulent velocity gradient. Table I lists all parameters. Three regimes are evaluated: Control ($D_{\text{bg}} = K = 0$), Moderate, and Extreme. A run is scored successful if the swarm’s best estimate comes within 0.3 m of $\vec{x}_f$ at any point in the horizon—used identically in Fig. 2 and Table II. Inertia/acceleration weights follow commonly-used PSO defaults [12]; velocity clamping is applied identically across all three benchmarked architectures so it cannot bias the comparison. All results below, including Fig. 2 and Table II, are generated directly by `pso_nuclear_sim.py` (Sec. IV); no entry is hand-entered.

Choice of N and $t_{\max}$. The domain spans $L \times 2R = 20$ m². $N = 25$ was not fixed a priori: Sec. III-B sweeps $N \in [5, 50]$ (Extreme profile) and finds success rising steeply through $N \approx 20$ –25 (34.2% $\rightarrow$ 86.0%), then flattening (88.8% at $N = 30$; 97.0% only at $N = 50$), while cumulative dose—a real hardware cost—grows linearly with N throughout (1815.8 $\rightarrow$ 19,249.1 Gy). $N = 25$ sits at this empirical knee: near-saturated coverage of Ω_{pipe} at $r_{\text{comm}} = 2.0$ m, without the compounding dose burden and $O(N^2)$ neighbor-graph cost (Eq. 3) of larger swarms for comparatively small gain. $t_{\max} = 150$ (15 s at $\Delta t = 0.1$ s) comfortably exceeds the Control/Moderate plateau near $t \approx 25$ –40 while still capturing the slower Extreme-profile tail through $t = 149$ (Fig. 2a), rather than truncating the regime where decentralization’s advantage is largest.



(a) Population-median convergence distance to the fissure across the three regimes (500 runs/regime).



(b) Agent pathways under Extreme with full turbulent shear (single representative run, 25 agents).

Fig. 2: Decentralized M-PSO convergence and trajectory behavior.

A. Convergence, Trajectory, and Baseline Comparison

A Monte Carlo campaign (500 runs/architecture/profile) compares Decentralized M-PSO against a Centralized Swarm (a single leader broadcasts a $\vec{g}^{\text{best}}$ that freezes—permanently, after leader latch-up—whenever $\chi_{\text{leader}}(t) = 0$) and a Canonical PSO (fully-connected, no drag term, no communication-degradation model). Performance is reported as Success Rate (fraction of 500 runs with $\|\vec{g} - \vec{x}_f\| \leq 0.3$ m, with n_{success} and a Wilson-score 95% CI), Convergence Time (successful runs only), and Cumulative Swarm Dose (Eq. 10, summed over all agents).

Control. The median distance (Fig. 2a) falls from 0.51 m to a plateau near 0.225 m by $t \approx 25$ —fastest of the three regimes, as expected. Yet Table II reports only 64.8% success for Decentralized M-PSO even here (Centralized 56.6%, Canonical 58.0%, overlapping CIs). This is not agent divergence: every failing run’s terminal distance was strictly between the 0.3 and 0.78 m threshold—classic premature-convergence (stagnation) [15]: once $\vec{p}_i \approx \vec{g}_i$, Eq. (2)’s cognitive/social terms vanish, and absent noise the swarm cannot escape a plateau just outside the success radius. This is a hyperparameter-dependent *floor* shared by all three architectures under Control, and is key to the Extreme-profile finding below. Centralized Swarm’s slower convergence (14.5 vs. 5.8 mean steps) reflects leader-broadcast latency even absent packet loss.

Moderate. The median distance plateaus higher and noisier, 0.32–0.34 m, explaining the drop in Decentralized success to 61.4%: Gaussian sensor corruption ($\sigma_\gamma^2 D(\vec{x})$ in $\vec{p}_i$) degrades final precision but not enough for the exploratory benefit seen next. Baselines fare far worse (Centralized 31.8%, roughly

TABLE II: Comparative Performance vs. Baseline Architectures (500 Monte Carlo Runs/Condition; Generated Directly by `pso_nuclear_sim.py`)

Profile	Architecture	Success (%)	n_{success}	95% CI	Mean Conv.	Median Conv.	Cum. Dose (Gy)
Control	Decentralized M-PSO	64.8	324/500	[60.5, 68.9]	5.8	3.5	0.0
	Centralized Swarm	56.6	283/500	[52.2, 60.9]	14.5	12.0	0.0
	Canonical PSO	58.0	290/500	[53.6, 62.2]	5.8	6.0	0.0
Moderate	Decentralized M-PSO	61.4	307/500	[57.1, 65.6]	12.0	5.0	1145.0 ± 98.8
	Centralized Swarm	31.8	159/500	[27.9, 36.0]	24.8	18.0	1406.9 ± 193.0
	Canonical PSO	40.8	204/500	[36.6, 45.2]	6.7	1.0	2974.7 ± 478.0
Extreme	Decentralized M-PSO	84.4	422/500	[81.0, 87.3]	34.0	23.0	9503.5 ± 754.6
	Centralized Swarm	1.4	7/500	[0.7, 2.9]	28.0	0.0	12649.2 ± 3484.3
	Canonical PSO	28.4	142/500	[24.6, 32.5]	4.8	1.0	24774.1 ± 5557.4

half its Control value; Canonical 40.8%): a packet drop affecting the Centralized leader freezes the entire swarm’s coordination, with no analogue in the neighborhood-restricted topology, where losing one agent’s edges leaves the rest of $\mathcal{G}(t)$ intact. Cumulative dose corroborates this: Decentralized’s faster convergence yields the lowest exposure (1145.0 Gy) vs. Canonical’s highest (2974.7 Gy), the latter persisting the full mission on many runs without communication to help it succeed.

Extreme. The median distance descends more slowly at first but keeps improving throughout, reaching the lowest final value of the three regimes (0.213 m at $t = 149$)—counter-intuitively, the larger sensor-noise variance here acts as a stochastic exploration source that escapes the same stagnation plateau seen under Control, at the cost of slower initial descent, higher dose, and the trajectory dispersion in Fig. 2b (turbulent forcing plus intermittent $\chi_i(t) = 0$ episodes). Radiological interference thus degrades convergence *speed* monotonically, but final precision is *non-monotonic*—Moderate is strictly worse than *either* Control or Extreme, reported honestly rather than smoothed over. Decentralized success *rises* to 84.4% (95% CI [81.0, 87.3]). Centralized Swarm instead collapses to 1.4% (7/500)—traced to early-mission leader latch-up: $\Lambda_i(t)$ crosses $\Lambda_{\text{fail}} = 300$ Gy well within the mission window near the fissure, and once this hits the leader, Eq. (11) permanently freezes the swarm-global estimate regardless of subsequent follower discoveries. Canonical PSO also degrades (28.4%) but, lacking any communication model to fail, is driven purely by sensor-noise corruption of its always-connected best—far less catastrophic than Centralized’s single point of failure. Despite Fig. 2b’s dispersion, no single trajectory’s failure compromises the outcome: since no agent holds a privileged role, a momentarily-connected subset persists each step to propagate an updated $\vec{g}_i(t)$ (Eq. 4), confirmed quantitatively here.

Taken together, these results substantiate decentralization as a measurable architectural necessity: the success-rate advantage over centralized coordination grows from a negligible Control-profile difference to a $> 60\times$ relative improvement (84.4% vs. 1.4%) under catastrophic adversity—precisely the regime where autonomous PWR inspection is most needed.

B. Sensitivity to Swarm Size and Communication Range

The results above fix $N = 25$ and $r_{\text{comm}} = 2.0$ m (Table I). To characterize robustness and generalization beyond the

TABLE III: Swarm-Size Sensitivity (Decentralized M-PSO, Extreme Profile)

N	Success (%)	Mean Conv. (steps)	Cum. Dose (Gy)
5	34.2	55.9	1815.8
10	54.2	49.8	3719.9
15	70.2	44.3	5625.1
20	82.2	36.2	7544.5
25	86.0	28.3	9515.8
30	88.8	25.5	11438.8
40	93.4	21.3	15348.9
50	97.0	19.7	19249.1

TABLE IV: Communication-Range Sensitivity (Extreme Profile, $N = 25$)

r_{comm} (m)	Decentralized M-PSO Succ. (%)	95% CI	Centralized Swarm Succ. (%)	95% CI
0.5	86.2	[82.9, 88.9]	1.6	[0.8, 3.1]
1.0	84.4	[81.0, 87.3]	0.6	[0.2, 1.7]
1.5	88.4	[85.3, 90.9]	0.8	[0.3, 2.0]
2.0	85.6	[82.3, 88.4]	1.0	[0.4, 2.3]
3.0	82.8	[79.2, 85.9]	1.2	[0.6, 2.6]

noise/radiation axis already varied across profiles, two further Monte Carlo sweeps (500 trials per condition, Extreme profile, identical success threshold) vary these system parameters directly.

Swarm size. Table III shows success rising steeply from 34.2% ($N = 5$) to 86.0% ($N = 25$), then flattening: marginal gain per agent falls from +20.0 points ($N:5 \rightarrow 10$) to +2.8 ($N:25 \rightarrow 30$) and +3.6 ($N:40 \rightarrow 50$), while cumulative dose grows linearly and unconditionally with N . This is the empirical basis for $N = 25$: below the knee, small swarms are frequently reduced to mutually-unreachable clusters by ordinary Bernoulli dropout, undersampling Ω_{pipe} ; above it, added agents mostly duplicate existing coverage while proportionally raising radiological exposure of the hardware—a real cost this paper’s dose model makes explicit.

Neighbor failure dynamics (communication range). Table IV addresses the concern that neighbors, not only a leader, can fail under the modeled radiological conditions—a concern this paper’s model already takes seriously, since $\chi_k(t)$ (Eq. 3) governs *every* agent’s transceiver identically and the latch-up of Eq. (11) can strike any agent. Shrinking r_{comm} from 3.0 to 0.5 m—forcing sparser neighborhoods and more frequent full isolation ($\mathcal{N}_i(t) = \{i\}$)—leaves Decentralized M-PSO flat within noise across the swept range (82.8–88.4%, overlapping CIs), whereas Centralized Swarm stays broken at every range (0.6–1.6%). This confirms the mechanism is architectural, not range-dependent: Decentralized M-PSO needs only that *some* non-empty connected subset exists each step (Eq. 4),

regardless of which agents. Centralized Swarm is insensitive to r_{comm} because its failure is not a range problem: the leader's own latch-up (Eq. 11) is fatal, and no range helps once the single aggregation point is disabled.

C. Comparison with Learning-Based Approaches

Recent deep reinforcement learning (DRL) and vision-based approaches show strong real-world results across locomotion, navigation, and manipulation [16], making DRL a natural comparison point. M-PSO is not proposed as outperforming DRL in unconstrained settings; two properties specific to sustained radiation favor a lightweight analytical controller here. **(i) SEU exposure.** DRL deployed on-agent typically needs dedicated NPU/GPU inference hardware; irradiation testing of edge-AI accelerators under realistic neutron flux has directly measured weight corruption and outright failure in this hardware class [17], whereas M-PSO's per-agent update (Eq. 2) is arithmetically trivial, executable on a minimal microcontroller with far smaller radiation-sensitive silicon and no dedicated accelerator. **(ii) Distribution shift.** DRL policies are trained under a fixed distribution and their robustness beyond it remains an open question [16]; M-PSO's update rule is a fixed analytical expression needing no training data for the encountered dose-rate regime, with explicit, inspectable stochastic terms ($r_1, r_2, \chi_i(t)$). This comparison is qualitative: benchmarking a trained DRL baseline under this same hazard model is future work (Sec. IV).

IV. CONCLUSION AND FUTURE WORK

This paper presented a computational framework for autonomous micro-agent inspection of PWR coolant loops under coupled turbulent-hydrodynamic and radiological adversity. Coupling a one-seventh power-law turbulent profile with non-linear drag, and a non-homogeneous gamma dose field with dual-phenomenon degradation (Gaussian sensor corruption, dose-scaled communication loss, and a cumulative-dose permanent-failure state), a reduced-order simulator produced every quantitative result in Sec. III. Canonical, drag-unaware PSO underperformed across all regimes; centralized leader-follower coordination proved structurally vulnerable to single-point collapse (1.4% success under catastrophic flux vs. 84.4% for the proposed M-PSO)—substantiating decentralization as a necessary architectural precondition, not an incremental refinement. Localization precision also depended non-monotonically on radiological severity (Sec. III-A), reported honestly rather than smoothed into a simpler narrative.

Limitations: (i) 2D, one-seventh power-law hydrodynamics rather than full 3D RANS/LES, omitting swirl and localized vortex shedding; (ii) homogeneous swarm (identical m, C_d, A , radiation hardness), neglecting manufacturing variability; (iii) thermal-hydraulic coupling neglected; (iv) N and r_{comm} are now empirically swept (Sec. III-B), but $(w, c_1, c_2, \Lambda_{\text{fail}})$ remain illustrative; (v) physical parameters are plausible but not validated against deployed platforms (MallARD [18], AVEXIS [19]), a prerequisite before hardware-in-the-loop claims.

Future work: 3D CFD (RANS/LES) co-simulation for vortex shedding and complex geometry; heterogeneous swarms with radiation-hardened relay nodes; Hardware-in-the-Loop validation with scaled flow-loop prototypes; a sweep of $(w, c_1, c_2, \Lambda_{\text{fail}})$ to further characterize the non-monotonic noise-precision relationship above; and a quantitative DRL baseline under this same hazard model to extend Sec. III-C beyond a qualitative comparison.

CODE AND DATA AVAILABILITY

The complete simulator source (pso_nuclear_sim.py), the sensitivity-sweep script (sensitivity_sweep.py) used for Table III and Table IV, the figure-generation script, and raw Monte Carlo output logs used to produce every quantitative result, table, and figure in this paper are permanently archived and publicly available on Zenodo at <https://doi.org/10.5281/zenodo.21703782>, so that all reported results may be independently reproduced or extended.

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